

Workforce Management System (WMS)

High-Level & Low-Level Design Document

1. Executive Summary

WMS is a web application built to help companies manage their workforce in one place. It handles employee records, attendance tracking, leave requests, project assignments, department information, and client data. Instead of using separate spreadsheets or tools for each of these, everything is available in a single secure system.

2. System Architecture

The system is split into three main parts that each handle a different job:

Part	Technology	What it does
Frontend	Angular 21	The screens and pages that users interact with
Backend	ASP.NET Core 8	Handles requests, applies business rules, and serves data
Database	Azure SQL	Stores all the data securely in the cloud

3. How the Code is Organised

The code is split into four layers so that each part only does one job. This makes it easier to fix bugs, add new features, and test things without breaking something else:

Layer	What it contains
Domain	The core business rules and data models. Nothing else depends on this.
Application	The logic that ties everything together — what happens when a user does something.
Infrastructure	The code that talks to the database and any external services.
API	The endpoints that the frontend calls, plus login and security checks.

4. How a New Employee Gets Added

When an admin adds a new employee, the request goes through several steps to make sure it is valid and saved correctly:

Step	Who/What	What happens
1	Admin	Fills in the employee form on the website
2	Angular (Frontend)	Checks the form for errors before sending it
3	JWT Middleware	Confirms the admin is logged in and has permission
4	Employee Controller	Receives the request and passes it to the right service
5	Employee Service	Runs the business logic and checks everything is correct
6	Repository	Prepares the data and sends it to the database
7	Azure SQL Database	Saves the new employee record

5. Database Structure

The Employee table is at the centre of the database. Everything else connects to it:

- Departments — which team or department the employee belongs to
- Attendance — records of when they checked in and out each day
- Leave — leave requests, whether they were approved, and remaining balance
- Projects — which projects they are assigned to and in what role
- Clients — linked indirectly through the projects they work on
- Login Info — their username, hashed password, and access role

6. Login and Security

When a user logs in, the system checks their password and, if correct, gives them a token. That token is attached to every request they make so the system knows who they are and what they are allowed to do. Here is how each security feature works:

Feature	How it works
JWT Token	A secure token is generated on login and checked on every request
Password Hashing	Passwords are scrambled before being saved — the original is never stored
Role-Based Access	Each user role (Admin, Manager, Employee) can only access certain pages

HTTPS	All data sent between the browser and server is encrypted
Input Validation	All data coming into the system is checked before it is processed

7. Technologies Used

Layer	Technology	Why we used it
Frontend	Angular 21	Popular framework for building fast, interactive web pages
Backend	ASP.NET Core 8	Reliable and fast framework for building APIs in C#
ORM	EF Core	Lets us work with the database using C# instead of raw SQL
Database	Azure SQL	Cloud-hosted database that handles backups and scaling automatically
Security	JWT + BCrypt	Standard tools for secure login and password storage

8. Conclusion

WMS brings together all the tools a company needs to manage its workforce into one clean, secure application. The way the code is structured makes it easy to maintain and extend over time. By using well-known technologies like Angular, ASP.NET Core, and Azure SQL, the system is built on a solid foundation that can grow with the organisation.